

Maitreyi Deepraj Dixit

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EDUCATION

Master of Science in Mechanical Engineering- Research- *Carnegie Mellon University, Pittsburgh, PA* (GPA: 4.0/4.0) May 2027

- Modern Control Theory, Artificial Intelligence/ Machine Learning, Robot Dynamics and Analysis, Robotics Software

Bachelor of Science in Mechanical Engineering- *Purdue University, West Lafayette, IN* (GPA: 3.6/4.0) May 2025

- Thermodynamics, Fluid Mechanics, Heat & Mass Transfer, Machine Design, Dynamics, Materials Science, Controls

WORK EXPERIENCE

Cummins Inc., Stoughton, WI

Design Engineering Co-op

Jun 2024- Dec 2024

- Automated CAD generation for the Twin Module program using Creo Pro/Program and Visual Basic, consolidating 15-20 configuration-specific workflows into one- improving release efficiency and assembly accuracy
- Designed and validated components using SimSolid power spectral density (PSD) and random broadband (RBM) analysis, ensuring components met structural and vibration requirements
- Defined tolerances, assembly specifications using DVA, GD&T, and Creo drawings to support manufacturable designs
- Evaluated bolt and canning material selections for new aftertreatment assemblies, supporting design reviews, regulatory compliance, and cross-functional coordination with suppliers and internal teams

RESEARCH EXPERIENCE

Biohybrid and Organic Robotics Group, Carnegie Mellon University

Research Assistant with Dr. Victoria Webster-Wood

Aug 2025- Present

- Designed and iterated a biomimetic soft “sea slug” robot, integrating actuator fabrication, custom PCB design, closed-loop control implementation, and neural-network models for adaptive locomotion
- Rapidly prototyped, experimentally evaluated designs, contributing to submissions to IEEE/RSJ IROS and IEEE ICRA
- Presented research at the Midwest Robotics Workshop and the NIH BRAIN Initiative Conference
- Developed a computer-vision pipeline in OpenCV to capture and analyze snake robot locomotion data, enabling automated motion characterization for bio-inspired robotics

Chortos Laboratory, Purdue University

Research Assistant with Dr. Alex Chortos

Jun 2023- May 2024

- Led undergraduate team and coordinated with graduate researchers, increasing experimental testing throughput by ~15 x
- Built robust Arduino-based test systems to characterize antioxidant polymer coated metal degradation
- Designed, fabricated, and validated custom apparatus using SolidWorks, 3D printing, and circuit assembly/soldering

PROJECTS

Soft Robotic Manta Ray- Senior Design Project, Role- Technical Lead

Jan 2025- May 2025

- Led development of a biomimetic silicone manta ray robot designed for 50m operating depth, designing molds in CAD, fabricating PCB with embedded microcontroller, and integrated electronics
- Validated pressure resistance, structural integrity, movement metrics, and electronics protection by component-level testing followed by 10-12m pool trials

Arduino Turret- Student Organization (Fictional Projects of Purdue), Role- Founder

Sep 2022- Jan 2024

- Founded and led development of an autonomous human-tracking turret, achieving ~80% targeting accuracy through integrated sensing and computer vision
- Designed Arduino-based circuitry and implemented computer-vision algorithms for real-time target detection and tracking

MSA Safety- Strategic Consulting Project, CMU Corporate Startup Lab

Jan 2026- May 2026

- Partnered with MSA Safety Product Team to evaluate connected gas-detection hardware, data capabilities, and product value to identify barriers to adoption versus legacy solutions
- Conducted technical & customer research across MSA, distributors and industrial users to develop product recommendations

INVOLVEMENT

- **Extracurricular Organizations:** Mechanical Engineering Graduate Student Organization (MS Liaison, Social Chair), Fictional Projects of Purdue (Founder), Pi Tau Sigma (ME Honor Society), CMU Abhinaya Dance, Women in Engineering Program (Tutor, Mentor), Alpha Theta Delta Design Fraternity
- **Project Teams:** American Society of Mechanical Engineers, Society of Women Engineers (Medical Devices)
- **Teaching:** Graduate TA- Dynamics, Mechanics Modeling with ANSYS, Mazes. Undergraduate TA- First Year Engineering

SKILLS, AWARDS AND CERTIFICATIONS

- **Programming & Controls:** Python, C/C++, MATLAB, Simulink, **Machine Learning:** PyTorch, OpenCV
- **CAD, Simulation & Hardware:** SolidWorks, PTC Creo, Siemens NX, Fusion 360, ANSYS, Gazebo, SimSolid, Arduino, KiCad, LTSpice, PCB Design, Rapid Prototyping, GD&T
- **Tools, Awards & Certifications:** LabVIEW, PTC Windchill, LaTeX; Dean’s List & Semester Honors, Andrew and Tracy Oxtoby Scholarship, Six Sigma Yellow Belt, Programming with Arduino Certification, CITI Research Training